# Statistical Evidence of Climate Change Through Long-Term Environmental Data

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## Problem Statement

Climate change is a dangerous and ever-changing phenomenon regarding our world's temperature. Patterns in emissions, sea levels, rainfall, and temperature are affected by climate change. Despite decades of data, there are unclear patterns in how different countries contribute and in the factors driving changes in the climate; we plan to use data to help solve these patterns.

Rising temperatures and emissions threaten agricultural operations, the ecosystem in both land and sea, and overall public health. Being able to predict climate change trends can help mitigate the damage caused by these issues. Farmers can plan for sudden changes in the climate, and specialists can help preserve certain ecosystems if given enough warning. Being able to resolve this issue before irreversible damage occurs is important.

## Research Questions & Hypothesis Matrix

| | Research Questions | Result |
|---|---|---|
| Q1 | **[Linear Regression]** Has average temperature increased? | Reject H0 |
| Q2 | **[Linear Regression]** Have extreme weather events increased? | $p=0.4947$ |
| Q3 | **[Linear Regression]** Which regions changed the most? | Reject H0 $p=0.020$ |
| Q4 | **[Levene's Test]** Has rainfall variability changed? | $p>0.05$ |
| Q5 | **[Pearson Correlation]** Can historical data support future planning? | Weak signal |
| Q6 | **[Pearson Correlation]** Does $CO_2$ correlate with temperature? | Reject H0 |
| Q7 | **[Independent T-Test]** Has sea-level rise accelerated? | $p=0.2411$ |
| Q8 | **[Pearson Correlation]** Relationship: temperature & sea-level rise? | $p=0.0622$ |
| Q9 | **[One-Way ANOVA]** Highest / lowest $CO_2$ contributors? | $p=0.0177$ |
| Q10 | **[Multiple Linear Regression]** Does $CO_2$ + Rainfall + Temp predict sea level? | $p=0.1347$ |

Key: If $p < 0.10$ Reject H0 ($\alpha=0.10$) If $p \geqq 0.10$ Fail to reject H0

## Dataset

- **Year —** Calendar year, 2000–2023
- **Country —** 15 nations across 6 continents
- **Avg Temperature (°C) —** Annual mean surface temperature
- **$CO_2$ Emissions (T/cap) —** Per-capita $CO_2$ in metric tons
- **Sea Level Rise (mm) —** Cumulative rise in millimetres
- **Rainfall (mm) —** Total annual precipitation
- **Population —** Country population for the year
- **Renewable Energy (%) —** Share from renewable sources
- **Extreme Weather Events —** Count of extreme events recorded
- **Forest Area (%) —** Land coverage by forests

## Conclusion

Climate change is a complex issue that affects every country differently. Using data from 15 countries over 24 years, we found that $CO_2$ emissions vary significantly between nations, with the UK being the highest contributor and Australia the lowest. Mexico showed the most notable warming trend, increasing by +0.43°C per year the only country where this trend was statistically significant. While a global temperature increase was observed, the diversity of countries in our dataset made it difficult to detect clear patterns overall. Many of our findings fell just outside our significance threshold of $\alpha = 0.10$, suggesting that with more data and a longer time range, stronger conclusions could be drawn. This research highlights the importance of country-specific climate policies and continued data collection to better understand and predict the effects of climate change before the damage becomes irreversible.

## Methodology

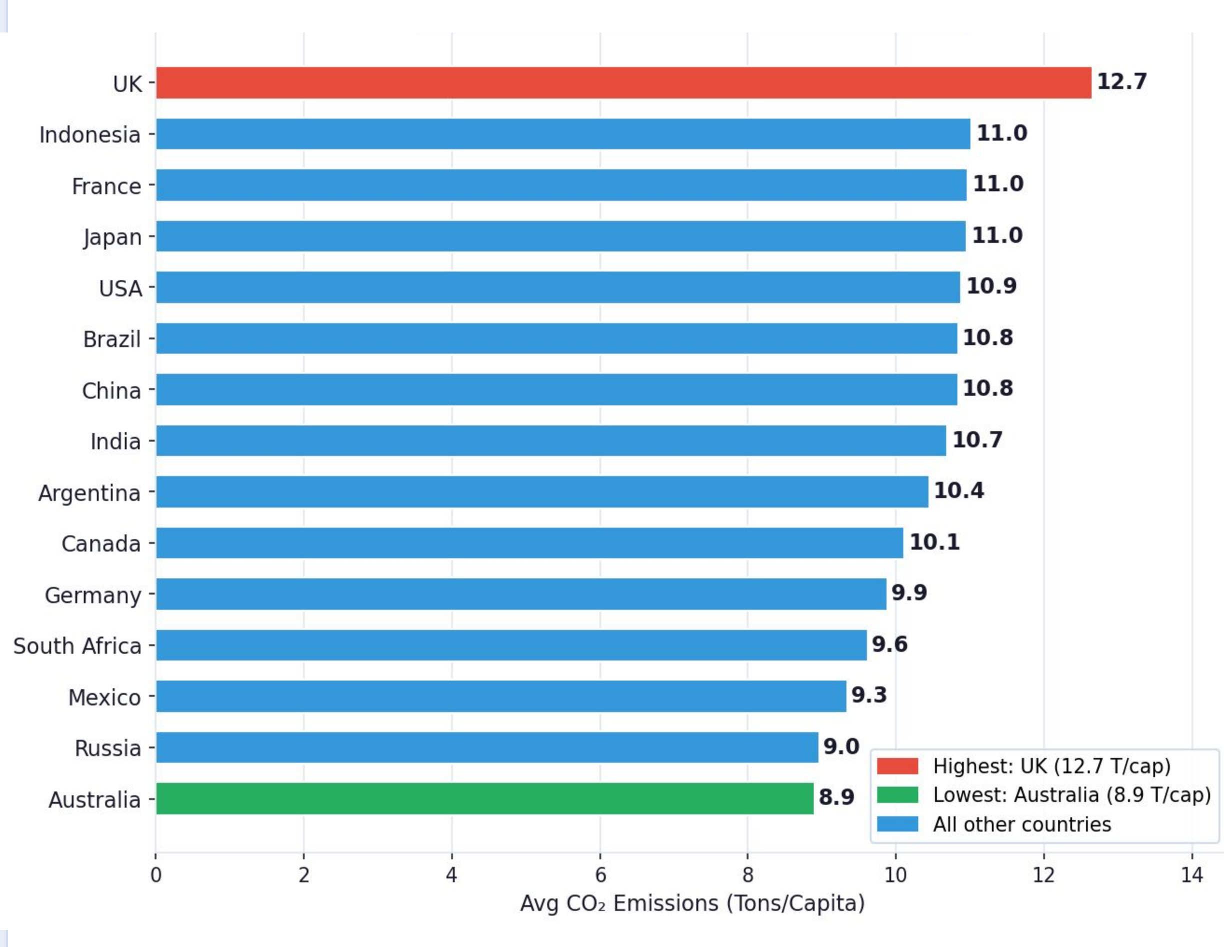
**Data Collection**

Kaggle climate-change-dataset (Bhadra Mohit). 1,000 country-year observations across 15 nations, 2000–2023, sourced from World Bank, NASA/NOAA, and IEA. As well as Data preprocessing which includes removing duplicates, handling missing values, and detecting outliers using statistical methods. After cleaning, descriptive statistics are computed to summarize key variables such as temperature, rainfall, and extreme weather frequency. Data analytic processes such as calculating the mean,median,mode,range and variance. As well as checking for Outliers using the IQR Method. Graphs including Box plots, Histograms, and bar charts.

**Statistical Tests Applied**

- **OLS Linear Regression:** Temporal trends in temperature & extreme events (Q1, Q2, Q3)
- **Pearson Correlation:** $CO_2$ vs temperature; renewable vs $CO_2$; temp vs sea level (Q5 ,Q6, Q8)
- **Levene's Test:** Equality of rainfall variance across time periods (Q4)
- **One-Way ANOVA:** $CO_2$ differences across countries; temp across emission tiers (Q9)
- **Independent T-Test (Welch):** Sea-level rise: early (≤2011) vs. recent (>2011) period (Q7)
- **Multiple Linear Regression:** $CO_2$ + Rainfall + Temperature → Sea Level Rise (Q10)
- **Segmentation Analysis:** $CO_2$ Emission Tiers & Composite Climate Risk Groups (Task 3)

## Key Findings (Charts & Tables)
### Largest CO2 emissions by country



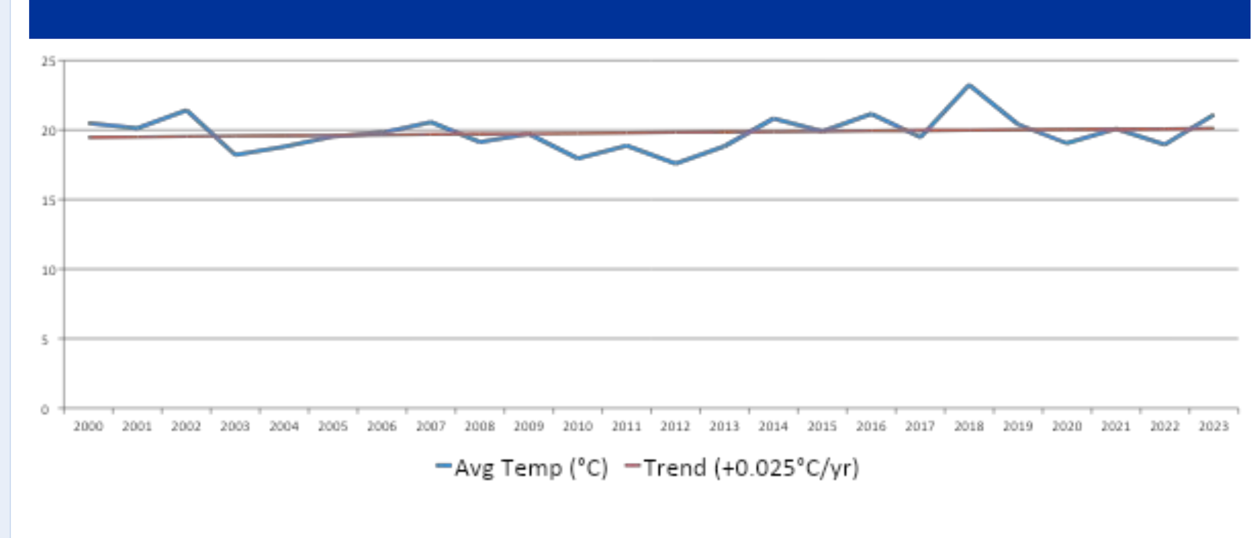

**+0.025°C/yr**
Global temp trend

**UK #1**
Highest $CO_2$ emitter
12.7 tons/capita

### CO2 Risk Profile

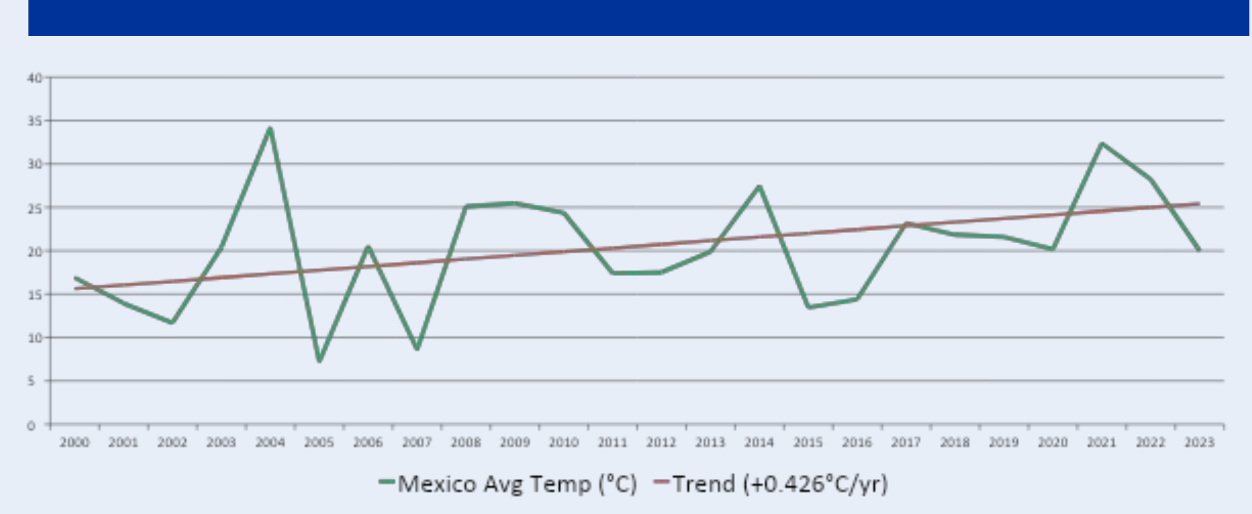
| Segment | $CO_2$ Range | Risk Profile |
|---|---|---|
| Low $CO_2$ Tier | < 7.4 T/cap | Moderate |
| Medium $CO_2$ Tier | 7.4–13.6 T/cap | Variable |
| High $CO_2$ Tier | > 13.6 T/cap | Higher |

### Average Temperature Over Time (All countries)




Average Increase = +0.025 Celsius per year across all 15 countries while Mexico had the largest increase at +0.43 Celsius per year

### Mexico Average temperature




## Future Work

1. Expand the dataset to 1960–2023 to gain greater statistical power for detecting slow-moving trends in sea-level rise and extreme events
2. Apply time-series models (ARIMA, Prophet) to each country independently to isolate within-country warming trends from cross-country panel noise
3. Develop a climate vulnerability index to guide targeted funding for highest-risk nations

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